Homework Submission Policy

- **Late submission:**
  - within 24 hours after its due will incur 20% penalty,
  - after 24 hours and within seven days of its due will incur 50% penalty, and
  - after seven days of its due will not be graded.

Note: One minute late is the same as 23 hours late.

Homework Assignment #3

Textbook Exercises (Based on the 9th edition textbook)

2.5-5. Let the moment-generating function …

2.5-6. Use the result of …

2.6-11. An airline always overbooks …

3.1-8. For each of the following …

3.2-4. Let $F(x)$ be the cdf of …

Due: 2:20pm, April 23, 2026

2.5-5. Let the moment-generating function $M(t)$ of X exist for $-h < t < h$. Consider the function $R(t) = \ln M(t)$. The first two derivatives of $R(t)$ are, respectively,

$$R'(t) = \frac{M'(t)}{M(t)} \qquad \text{and} \qquad R''(t) = \frac{M(t)M''(t) - [M'(t)]^2}{[M(t)]^2}.$$

Setting $t = 0$, show that

(a) $\mu = R'(0)$.

(b) $\sigma^2 = R''(0)$.

2.5-6. Use the result of Exercise 2.5-5 to find the mean and variance of the

(a) Bernoulli distribution.

(b) Binomial distribution.

(c) Geometric distribution.

(d) Negative binomial distribution.

2.6-11. An airline always overbooks if possible. A particular plane has 95 seats on a flight in which a ticket sells for \$300. The airline sells 100 such tickets for this flight.

(a) If the probability of an individual not showing up is 0.05, assuming independence, what is the probability that the airline can accommodate all the passengers who do show up?

(b) If the airline must return the \$300 price plus a penalty of \$400 to each passenger that cannot get on the flight, what is the expected payout (penalty plus ticket refund) that the airline will pay?

3.1-8. For each of the following functions, **(i)** find the constant c so that $f(x)$ is a pdf of a random variable X, **(ii)** find the cdf, $F(x) = P(X \le x)$, **(iii)** sketch graphs of the pdf $f(x)$ and the distribution function $F(x)$, and **(iv)** find μ and σ^2:

(a) $f(x) = x^3/4, \qquad 0 < x < c.$

(b) $f(x) = (3/16)x^2, \qquad -c < x < c.$

(c) $f(x) = c/\sqrt{x}, \qquad 0 < x < 1.$ Is this pdf bounded?

3.2-4. Let $F(x)$ be the cdf of the continuous-type random variable X, and assume that $F(x) = 0$ for $x \le 0$ and $0 < F(x) < 1$ for $0 < x$. Prove that if

$$P(X > x + y \mid X > x) = P(X > y),$$

then

$$F(x) = 1 - e^{-\lambda x}, \qquad 0 < x.$$

HINT: Show that $g(x) = 1 - F(x)$ satisfies the functional equation

$$g(x + y) = g(x)g(y),$$

which implies that $g(x) = a^{cx}$.